The World as a Graph Database for Distributed Robotic Intelligence and Autonomous Optimization

Ilakkuvaselvi Manoharan

Independent Researcher, Naperville, Illinois, USA
Correspondence: ilakkmanoharan@gmail.com

November 1, 2026

Preprint, not peer reviewed

© 2026 Ilakkuvaselvi Manoharan. Licensed under CC BY 4.0.

Perspective manuscript prepared for submission to Frontiers in Robotics and AI (Multi-Robot Systems)

Main text: 2237 words (excluding abstract, headings, declarations, and references) | Abstract: 192 words | Figures: 0 | Tables: 0

Abstract

Robotic systems working in shared physical environments repeatedly acquire information about objects, resources, and changing conditions. Exchanging that information may reduce redundant work, but it also creates costs and risks associated with communication, stale evidence, and conflicting actions. This Perspective proposes a federated world graph connecting specialized onboard capabilities with task-relevant shared knowledge through local gateways that may also provide charging. The graph represents partial observations and their provenance rather than a complete digital copy of reality. Robots retrieve bounded, authorized views while retaining local control and explicitly constrained behavior during disconnection. Building on shared robotic knowledge, dynamic scene graphs, cloud robotics, and task allocation, the paper argues for evaluating evidence freshness, resource ownership, and charging access together. A hypothetical tomato-monitoring scenario illustrates the distinction between measurements, inferences, assignments, and verified actions. The paper then proposes controlled comparisons against independent robots, message sharing, and centralized graph services, with total energy per verified successful task as a primary outcome. No implementation, experimental results, or demonstrated energy savings are claimed. The resulting agenda makes the broad vision of distributed intelligence and autonomous optimization testable through bounded deployments, explicit knowledge contracts, and system-level resource accounting.

Keywords: multi-robot systems, knowledge graphs, distributed robotics, local autonomy, energy efficiency, robotic agriculture, autonomous optimization

1 Introduction

Robots operating in farms, warehouses, and public infrastructure must act on changing physical conditions while managing limited energy, computation, and connectivity. A robot can observe its immediate surroundings, but another robot may already have measured the same soil patch, detected a blocked route, or discovered an unavailable charging station. Repeating these observations can consume resources without improving decisions. Sharing information introduces its own costs, including transmission, storage, reconciliation, and the risk of acting on stale evidence.

I propose “the world as a graph database” as an organizing perspective for this trade-off. Here, “world” means the task-relevant environment represented through incomplete observations. It does not mean a complete digital copy of physical reality. The proposed system connects specialized onboard capabilities to a federated, evolving graph through local gateways that may also provide charging. Robots retrieve bounded, authorized views and contribute observations whose origins and validity remain explicit.

The central argument is that the unit of coordination should include the physical task, the evidence needed to perform it, and the energy required to obtain that evidence. The proposed contribution is an architectural synthesis and evaluation agenda, rather than a new graph database, a validated control algorithm, or a claim of priority for shared robotic knowledge. No implementation or experimental results are reported. Agricultural monitoring provides a concrete running example, while delivery, infrastructure maintenance, and inventory management illustrate the wider scope.

2 Foundations and the proposed perspective

RoboEarth established a design and implementation for sharing knowledge between robots (Waibel et al., 2011). KnowRob linked robot control to knowledge processing, drawing on information computed from the robot’s internal data structures, perception, and external sources (Tenorth and Beetz, 2013). Cloud robotics research examined shared data, computation, and collective learning as

resources beyond an individual machine (Kehoe et al., 2015). These precedents establish knowledge exchange as a foundation of the present proposal rather than as its contribution.

Representing an environment as a graph also has direct precedents. Kimera combines metric and semantic information in three-dimensional dynamic scene graphs (Rosinol et al., 2021). Such representations motivate linking objects and places to the evidence supporting them. They do not, by themselves, establish that a shared graph will reduce fleet energy consumption or coordinate charging effectively.

Multi-robot task allocation addresses which robot should execute which task and provides a formal basis for analyzing coordination choices (Gerkey and Matarić, 2004). A graph supplies structured inputs to such decisions; it does not replace an allocation algorithm. Likewise, a graph database stores and retrieves relationships; a world model must additionally support prediction; and causal reasoning requires assumptions and evidence about interventions. These functions should remain distinguishable.

This Perspective advocates evaluating knowledge freshness, task allocation, charging access, and local autonomy together. Its distinctive emphasis is the operational contract between them: what evidence a task requires, how current that evidence must be, who may access it, and what a robot should do when the contract cannot be met. The originality of this synthesis should be judged against these established research traditions, rather than against the broad idea of connecting robots.

3 A federated graph of observations and actions

The graph contains entities such as robots, plants, soil zones, paths, tools, chargers, and tasks. Relations describe location, accessibility, resource requirements, observations, and completed actions. A plant may be located in a bed, associated with a soil sensor, and linked to an inspection task. Graph vertices are digital entities; physical gateways are computing and communication infrastructure. Using separate terms avoids confusing the representation with its deployment.

Each observation should contain an identifier, subject, measured property, value, unit, spatial reference, event time, ingestion time, source, and an uncertainty description appropriate to the measurement. An inferred disease label additionally requires the model version and a link to its supporting image. Records of derivation and responsible agents can follow the concepts defined in the W3C PROV-O provenance ontology (Lebo et al., 2013). Provenance makes an assertion traceable; it does not establish its correctness.

Observations, inferences, predictions, and action records remain separate types. A moisture measurement is evidence. A recommendation to irrigate is an inference under a policy. A predicted moisture trajectory is a model output. An irrigation completion record must be supported by actuator feedback or another defined verification mechanism. This separation prevents a proposed action from being mistaken for an accomplished one.

Regional operators maintain their own graph partitions and expose selected views through agreed schemas. A robot requests entities within a working area and relationships needed for its current task. Geometry, images, and high-frequency telemetry can remain in specialized stores, with references in the graph. This limits the cost of forcing every data modality into one representation. Stable identifiers, coordinate transforms, units, and schema versions are prerequisites for meaningful exchange.

The system should retain competing observations when they cannot be reconciled safely. A newer timestamp alone is insufficient to decide which sensor is correct. Reconciliation requires domain rules covering calibration, spatial coverage, uncertainty, and dependence between sources. Missing

information remains unknown. The proposal does not assume that thoughts or emotions can be directly observed; any permitted behavioral inference would require separate validation and explicit uncertainty.

4 Local autonomy and shared infrastructure

A robot maintains onboard perception, motion control, obstacle response, and a task cache. Its immediate safety functions must operate without a graph query. A local gateway authenticates requests, serves regional information, receives observations, and coordinates shared resources. Federation services exchange authorized summaries across gateways and support historical analysis. Training and expensive planning may use remote computation, but execution cannot assume a continuous connection.

Charging and information exchange can be colocated, allowing synchronization during an energy stop. Colocation is a deployment option rather than a requirement: a robot must also be able to communicate during a task when connectivity is available. Charger records should specify physical compatibility, capacity, estimated availability, and reservation state. A recorded charger location is insufficient evidence that a robot can reach or use it.

Historical observations can tolerate delayed replication when applications understand their age. Conflict-free replicated data types provide convergence guarantees for suitably designed data types (Shapiro et al., 2011), but convergence does not resolve conflicting interpretations of physical evidence. Exclusive task assignments and charger reservations require a separate ownership mechanism. A possible implementation uses a regional authority, expiring leases, and fencing tokens that resource controllers validate; lease expiry alone cannot prevent a disconnected robot from acting on an obsolete assignment.

During disconnection, a robot continues only actions allowed by its cached task contract. For example, it may finish a permitted inspection while postponing irrigation that depends on shared ownership. If it cannot establish that a resource is available, it defers the conflicting action. Rerouting requires a reachable alternative gateway, adequate energy, and authorization; it is not an automatic consequence of federation. Emergency behavior remains local and takes priority over synchronization.

Onboard computation also shapes the energy budget. Specialized models can support bounded functions such as plant detection or terrain classification. Distillation offers a route for transferring learned behavior into smaller models (Hinton et al., 2015), and domain randomization illustrates one approach to transferring visual models from simulation (Tobin et al., 2017). Neither guarantees lower system energy or safe field performance. Model selection requires measurements on the intended hardware, including sensing, actuation, and communication costs. General models may remain useful for occasional planning or exception handling.

5 From domain intelligence to coordinated action

The broader vision motivating this proposal spans behavioral, spatial, mobility, environmental, biological, agricultural, infrastructure, energy, inventory, predictive, and collective intelligence. In an implementable architecture, these are task-oriented services over overlapping evidence, rather than independent kinds of intelligence automatically produced by graph connectivity. Spatial services estimate routes and traversability; agricultural services interpret plant and soil observations; inventory services connect stock, demand, and replenishment tasks. Each service needs its own models, validation criteria, and access boundaries.

A graph can connect these services without granting every robot access to every record. A gardening robot may retrieve soil measurements and path obstructions, while a delivery robot retrieves destinations and charger availability. Neither needs personal identity records for those tasks. Shared concepts such as location and time enable interoperability, but application-specific meaning must remain explicit. A soil disease prediction cannot be reused as a general biological diagnosis merely because both are graph assertions.

Collective intelligence, in this proposal, means that one robot's supported observation can improve another robot's decision. Optimization requires an explicit objective and constraints, and automation requires an authorized executable action. For example, an allocator might prioritize urgent inspections subject to battery reserve, water availability, route feasibility, and exclusive resource access. A graph makes relevant relationships available; an optimizer and controller still determine and execute the plan.

6 An illustrative tomato-monitoring scenario

Consider a proposed deployment, referred to here as TOMO, in a bounded tomato-growing area with two mobile robots, fixed soil sensors, and a gateway beside a compatible charger. This is a hypothetical design example, not a description of an operating prototype. The graph links plants to beds, soil zones, images, irrigation equipment, and inspection history. Monitoring and watering are separate capabilities; seed sowing would require additional manipulation hardware and task-specific verification.

The first robot records a soil measurement and an image of a plant. An interpretation service proposes an inspection or watering task according to a policy that specifies acceptable measurement age and confidence. The second robot retrieves the evidence and checks its own capabilities, battery reserve, and task ownership. If watering is permitted, it obtains a valid assignment before activating the relevant equipment. It records commanded water volume separately from verified delivery and schedules a later measurement when appropriate.

If the moisture estimate is stale or disagrees with another sensor, the robot remeasures or requests review. If the charger is unavailable, it revises its route while preserving the configured reserve. If communication fails before task ownership is established, it postpones the shared action. These branches make the knowledge contract observable and testable. A human interface displays the evidence, uncertainty, outstanding tasks, and reasons for deferral rather than presenting every inferred state as fact.

7 A falsifiable evaluation agenda

The first hypothesis is that task-relevant sharing reduces total energy per verified successful task when reusable observations are sufficiently frequent. The second is that selecting evidence by relevance and freshness reduces communication relative to indiscriminate sharing while preserving task quality. The third is that local autonomy and explicit ownership rules improve useful operation during outages without increasing conflicting physical actions. These are proposed hypotheses, not established outcomes.

A future evaluation should compare independent robots, robots sharing conventional messages or a common database, a centralized graph service, and the proposed federated graph. The message-sharing comparator is essential: it tests whether relational queries add value beyond communication itself. All conditions should use the same robot capabilities, task workloads, perception models, and safety rules. Graph-specific engineering effort and query overhead should be reported alongside operational metrics.

Experiments should vary fleet size, observation reuse, network latency, outage duration, environmental change, sensor error, and charger contention. Predefined workloads should include fresh evidence, contradictory measurements, lost acknowledgments, expired reservations, and unavailable gateways. A staged program can begin in simulation and proceed to a bounded field trial only after the control and ownership mechanisms are verified. Simulation cannot establish field reliability on its own.

Primary outcomes should include total operating energy per verified successful task, task completion and quality, and conflicting action attempts. Energy accounting must include robot sensing and motion, onboard computation, network equipment, gateways, attributable remote computation, and charging losses. Both marginal energy and an allocation of standing infrastructure consumption should be disclosed. Training energy should be reported separately with its assumed reuse horizon. Comparing inference power alone would not test the system-level claim.

Secondary outcomes include transferred bytes, evidence age at decision time, query latency, repeated observations, charger waiting, intervention frequency, and recovery time. Ablations should remove freshness filters, relevance filtering, charging coordination, and ownership enforcement individually. Repeated trials with paired workloads, reported uncertainty intervals, and a predeclared analysis plan should distinguish systematic effects from variability. The proposal would be weakened if message sharing matches its benefits at lower cost, or if reconciliation and infrastructure overhead outweigh avoided work.

8 Discussion

The architecture depends on interoperability and trustworthy evidence. Entity matching can merge distinct objects, coordinate errors can associate observations with the wrong plant, and compromised contributors can poison shared knowledge. Authenticated provenance, validation, revocation, retention limits, and audit records are therefore design requirements. Query authorization should consider the task and purpose as well as the requesting identity, and apply to derived results and caches. Cached access requires an explicit expiry policy because immediate revocation cannot be guaranteed during disconnection.

The proposal also has limits of scope. It does not establish global consistency, a universal ontology, reliable causal inference, or planetary scalability. High-fidelity representations across biological and urban scales would require different resolutions, institutions, and validation practices. Traffic control, public surveillance, and law enforcement introduce distinct legal and safety requirements and are outside the proposed agricultural evaluation. Transfer to those domains requires independent assessment.

Energy efficiency remains conditional. Specialized models may fail outside their operating conditions, gateways consume standing power, synchronization can increase network load, and traveling to a charger can erase computational savings. Solar-supported charging requires site-specific energy and storage analysis. A successful architecture must expose these costs rather than assume that distributed computation eliminates large-scale infrastructure needs.

The world-as-graph perspective is useful when it makes the relationship between evidence and action explicit. Its research value lies in testing whether robots can coordinate through partial, traceable knowledge while retaining safe local behavior and accounting for total resource use. A bounded agricultural study offers a practical starting point. Broader autonomous optimization should follow demonstrated benefits, interoperable interfaces, and validated operating limits.

Data availability statement

No datasets were generated or analyzed for this conceptual Perspective.

Author contributions

IM: Conceptualization, Writing – original draft, Writing – review & editing.

Funding

The author declares that no financial support was received for the research and/or publication of this article.

Conflict of interest

The author declares that the research was conducted in the absence of any commercial or financial relationships that could be construed as a potential conflict of interest.

Acknowledgments

Generative AI assistance was used to organize, draft, and edit this manuscript and to assist with literature discovery. Tool: OpenAI ChatGPT with Codex; model: GPT-6; source: https://chatgpt.com. AI was not used to generate experimental data, and no empirical results are reported. Grok Bot was used for copy-editing and reference checking.

References

Gerkey, B. P., and Matarić, M. J. (2004). A formal analysis and taxonomy of task allocation in multi-robot systems. International Journal of Robotics Research 23, 939–954. doi: 10.1177/0278364904045564.

Hinton, G., Vinyals, O., and Dean, J. (2015). Distilling the knowledge in a neural network. arXiv [Preprint]. arXiv:1503.02531. doi: 10.48550/arXiv.1503.02531.

Kehoe, B., Patil, S., Abbeel, P., and Goldberg, K. (2015). A survey of research on cloud robotics and automation. IEEE Transactions on Automation Science and Engineering 12, 398–409. doi: 10.1109/TASE.2014.2376492.

Lebo, T., Sahoo, S., and McGuinness, D. (eds.) (2013). PROV-O: The PROV Ontology. W3C Recommendation, 30 April 2013. Available at: https://www.w3.org/TR/prov-o/ (Accessed November 1, 2026).

Rosinol, A., Violette, A., Abate, M., Hughes, N., Chang, Y., Shi, J., et al. (2021). Kimera: From SLAM to spatial perception with 3D dynamic scene graphs. International Journal of Robotics Research 40, 1510–1546. doi: 10.1177/02783649211056674.

Shapiro, M., Preguiça, N., Baquero, C., and Zawirski, M. (2011). Conflict-free replicated data types. In Stabilization, Safety, and Security of Distributed Systems (SSS 2011), Lecture Notes in Computer Science, Vol. 6976 (Berlin, Heidelberg: Springer), 386–400. doi: 10.1007/978-3-642-24550-3_29.

Tenorth, M., and Beetz, M. (2013). KnowRob: A knowledge processing infrastructure for cognition-enabled robots. International Journal of Robotics Research 32, 566–590. doi: 10.1177/0278364913481635.

Tobin, J., Fong, R., Ray, A., Schneider, J., Zaremba, W., and Abbeel, P. (2017). Domain randomization for transferring deep neural networks from simulation to the real world. In 2017 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS) (Vancouver, BC: IEEE), 23–30. doi: 10.1109/IROS.2017.8202133.

Waibel, M., Beetz, M., Civera, J., D'Andrea, R., Elfring, J., Gálvez-López, D., et al. (2011). RoboEarth—A world wide web for robots. IEEE Robotics & Automation Magazine 18, 69–82. doi: 10.1109/MRA.2011.941632.